

Supply_Chain Performance Report 2024 (ID: doc-syn-10)

1. Executive Summary & Production Efficiency Analysis

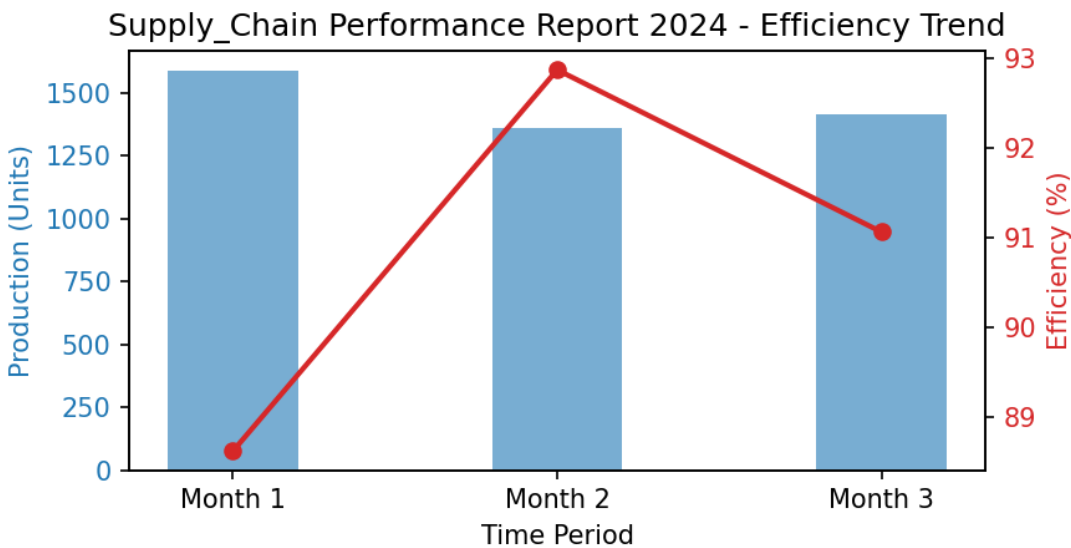
During the operational assessment of Supply_Chain Performance Report 2024, production efficiency experienced variance. The primary root cause for downtime in Q2 was unexpected hydraulic failure in Pump-04, whereas in Q4 the downtime was driven by supply chain shortages of raw titanium alloy.

2. Manufacturing Metrics & Output Table

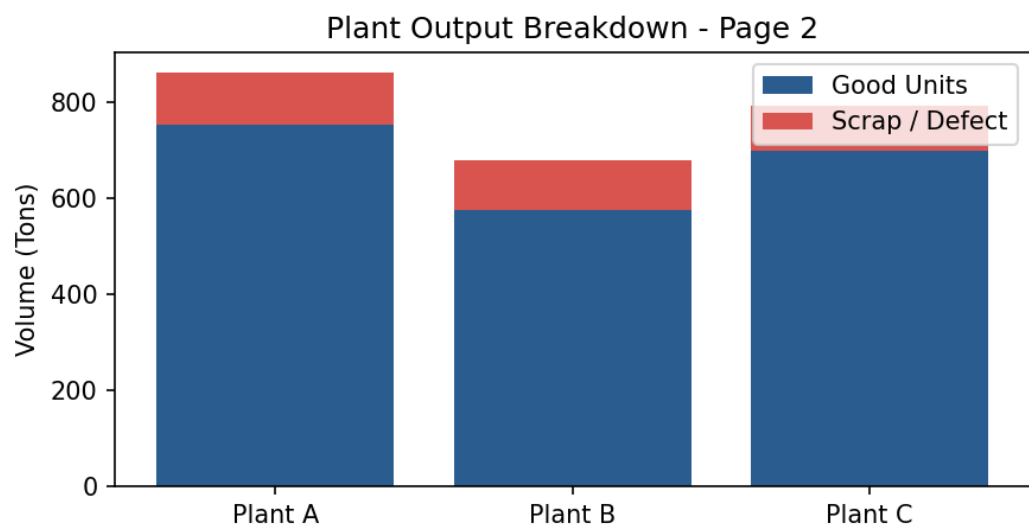
Metric Category	Quarter Baseline	Actual Output*	Efficiency Rate (%)
Line Alpha (Heavy)	1200 Units/Day	1416 Units/Day	85.8%
Line Beta (Light)	800 Units/Day	991 Units/Day	77.2%
Total Operational Output	2000 Units/Day	2407 Units/Day	81.5%

**Note: Actual output includes unadjusted scrap margin of 3.5% for Units/Day.*

3. Dual-Axis Production vs Efficiency Trend



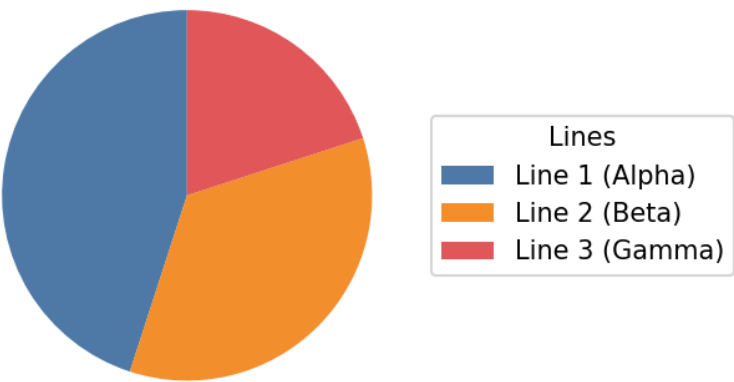
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ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

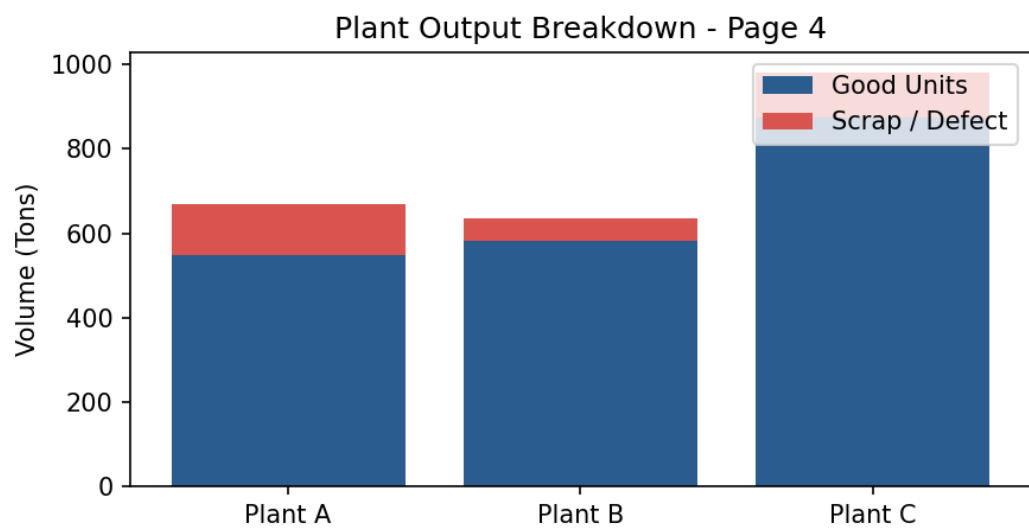
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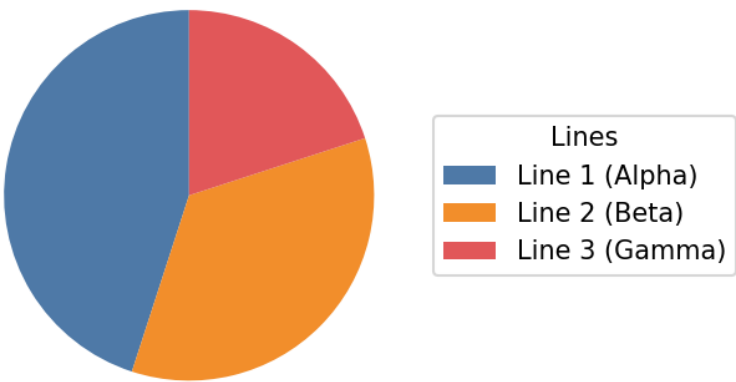
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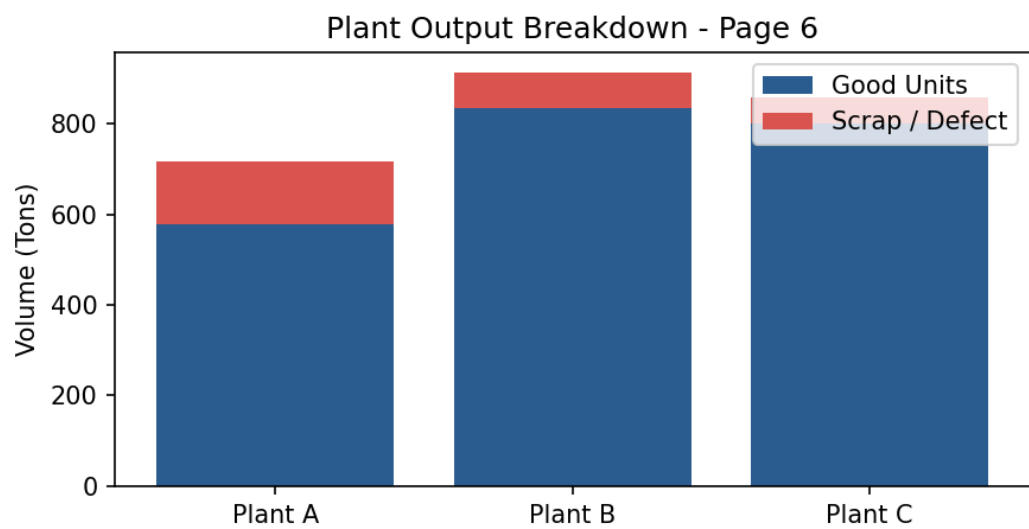
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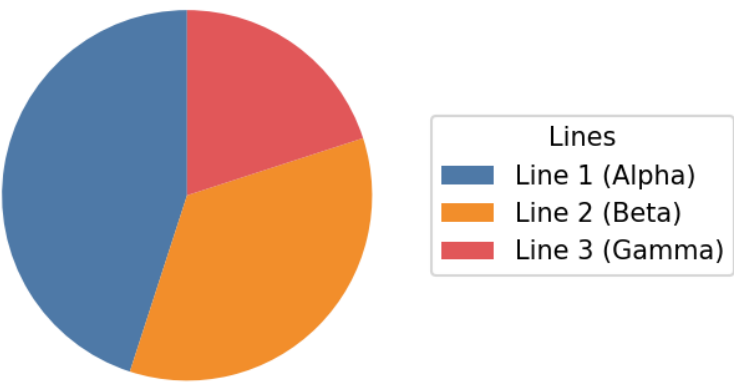
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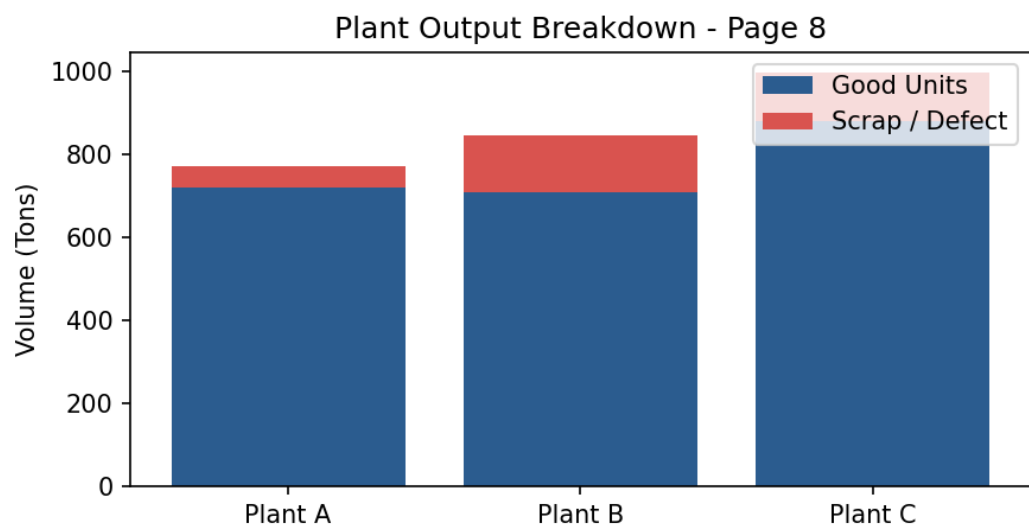
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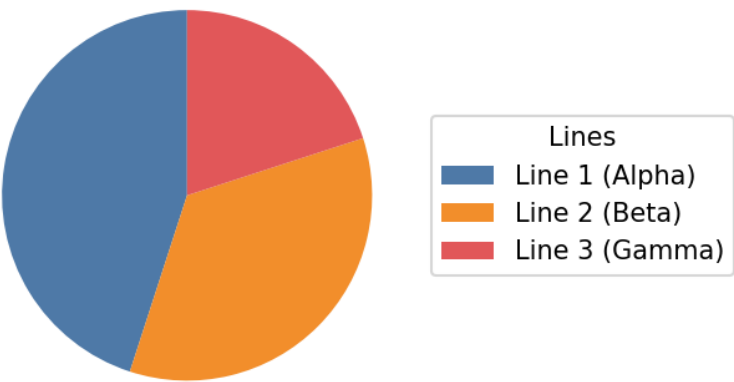
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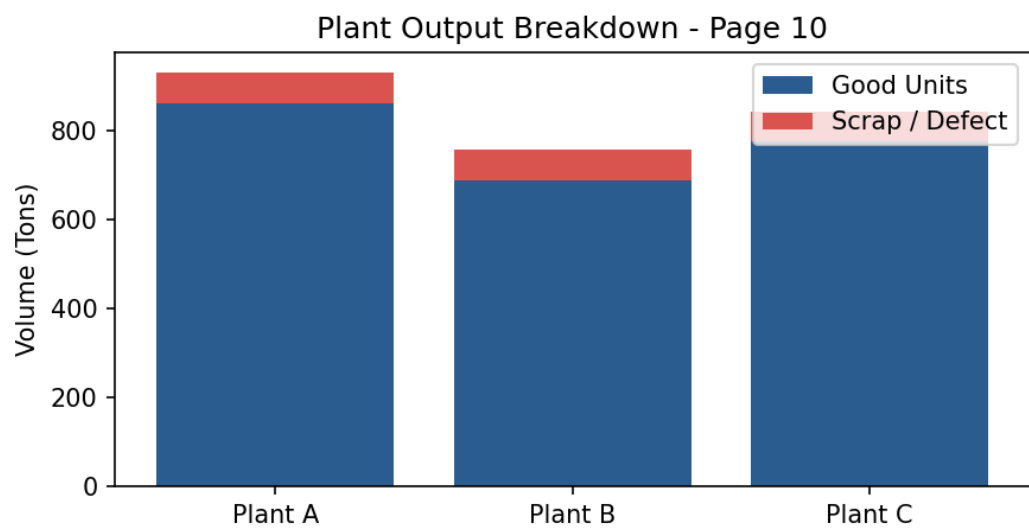
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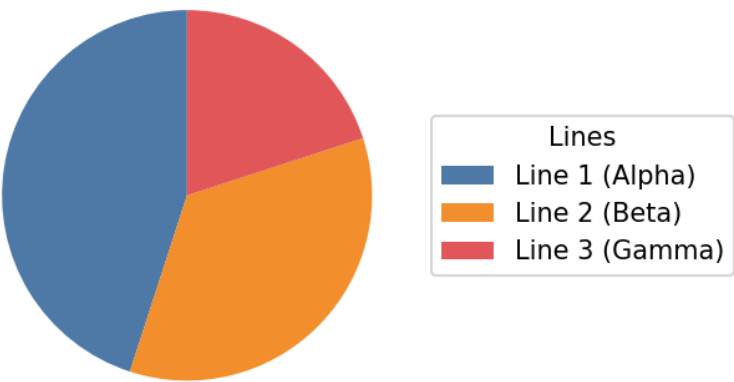
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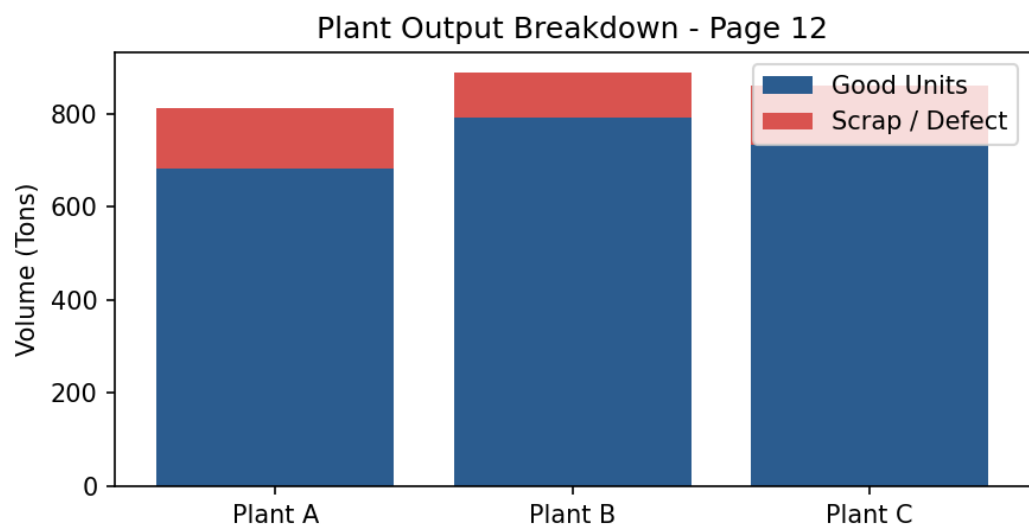
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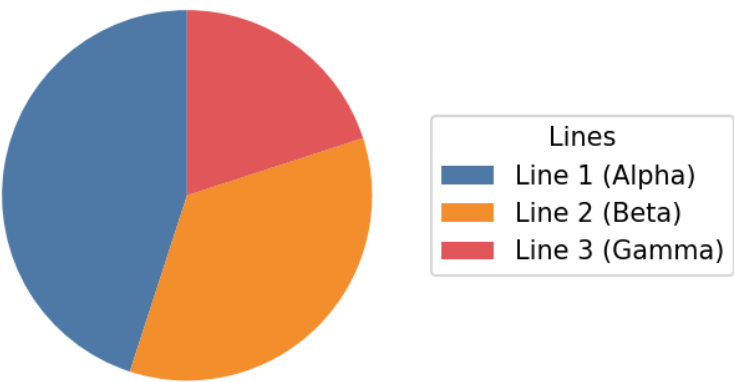
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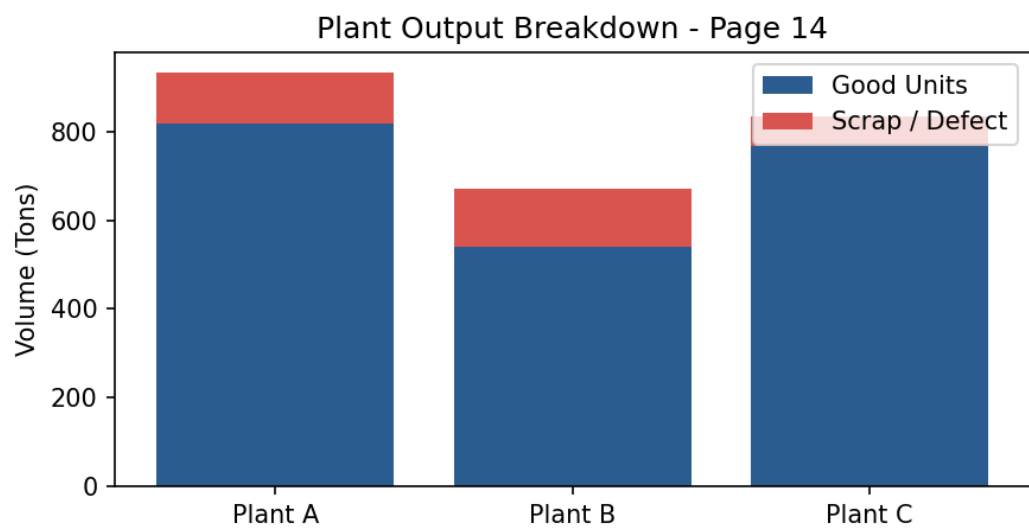
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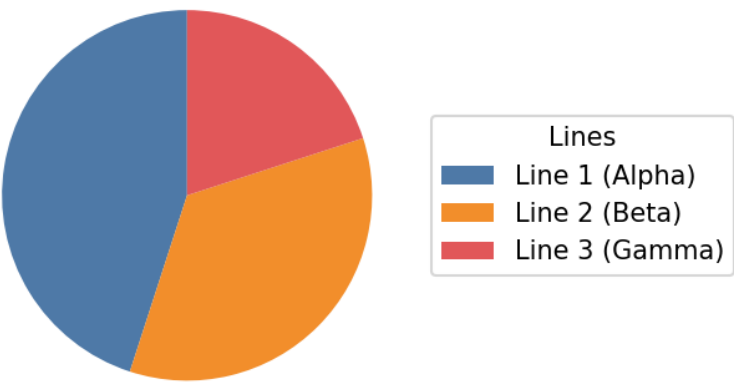
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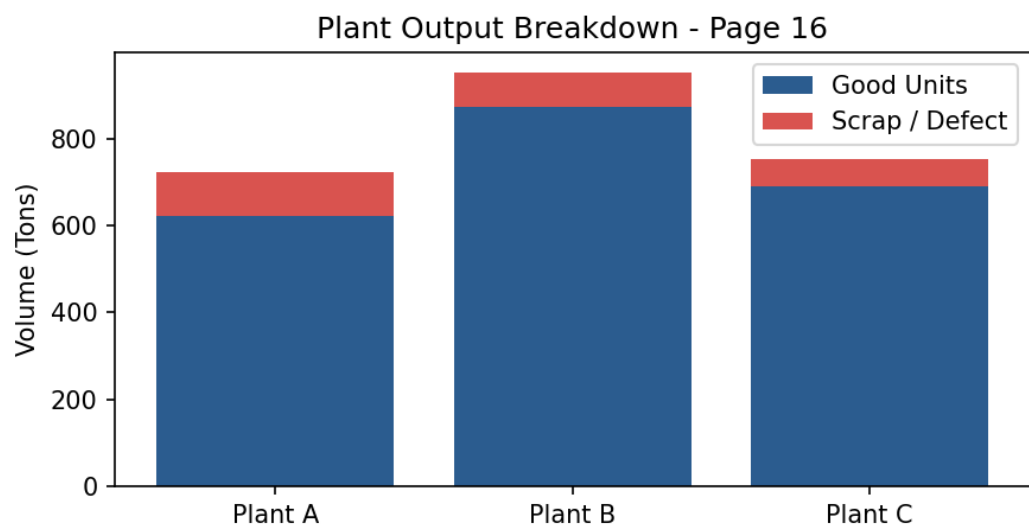
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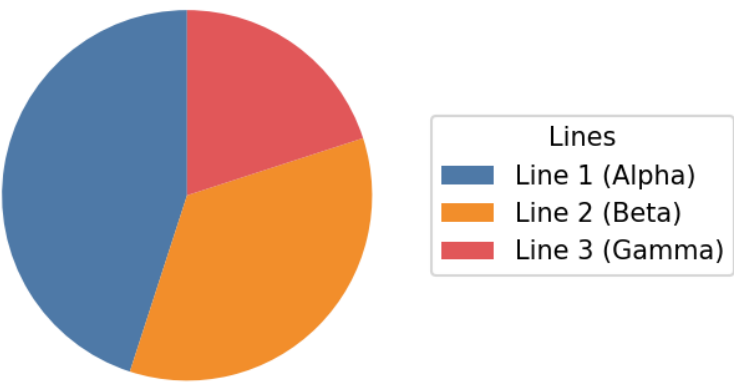
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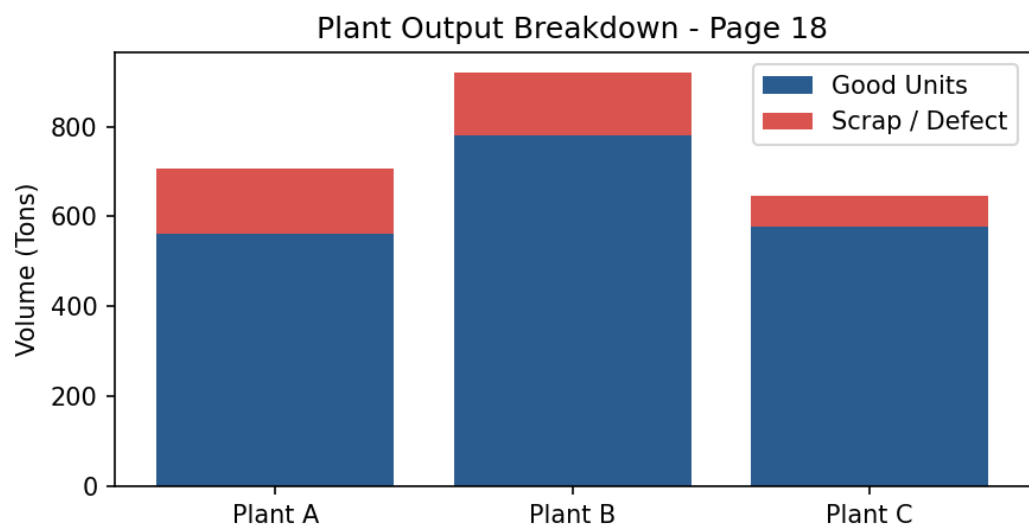
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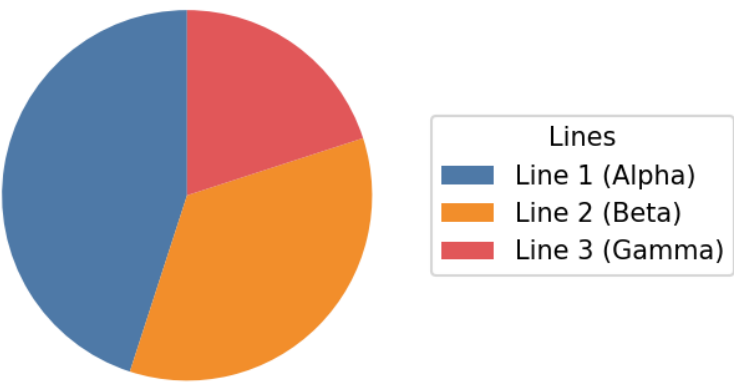
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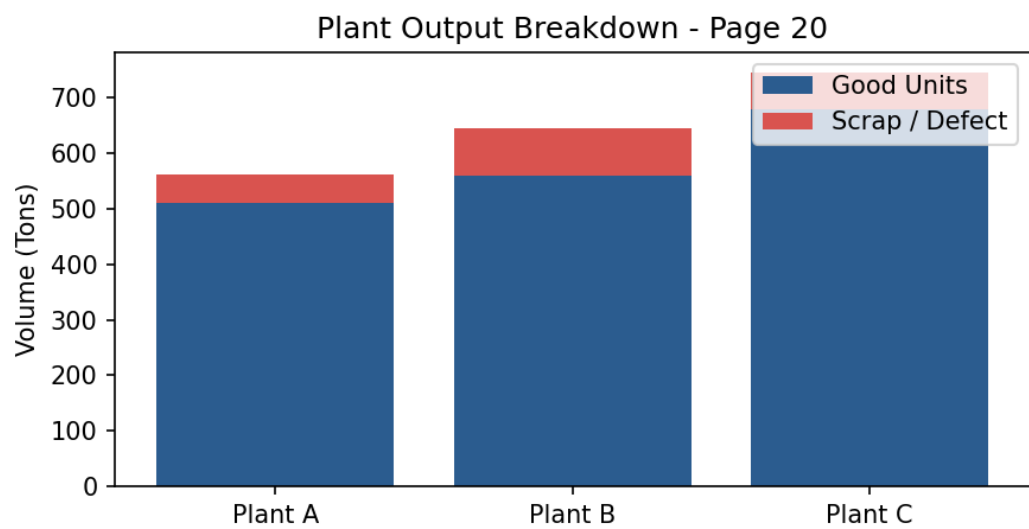
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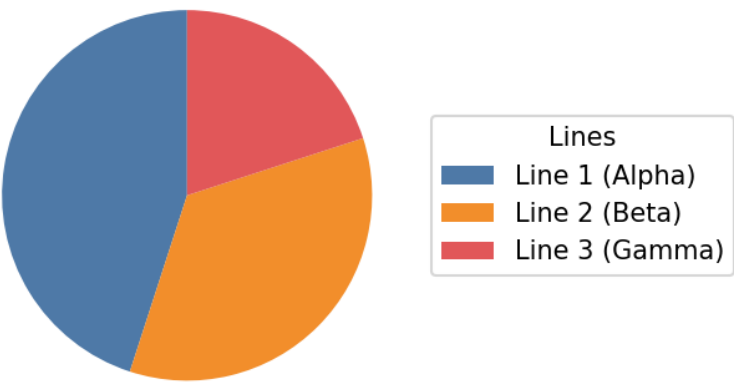
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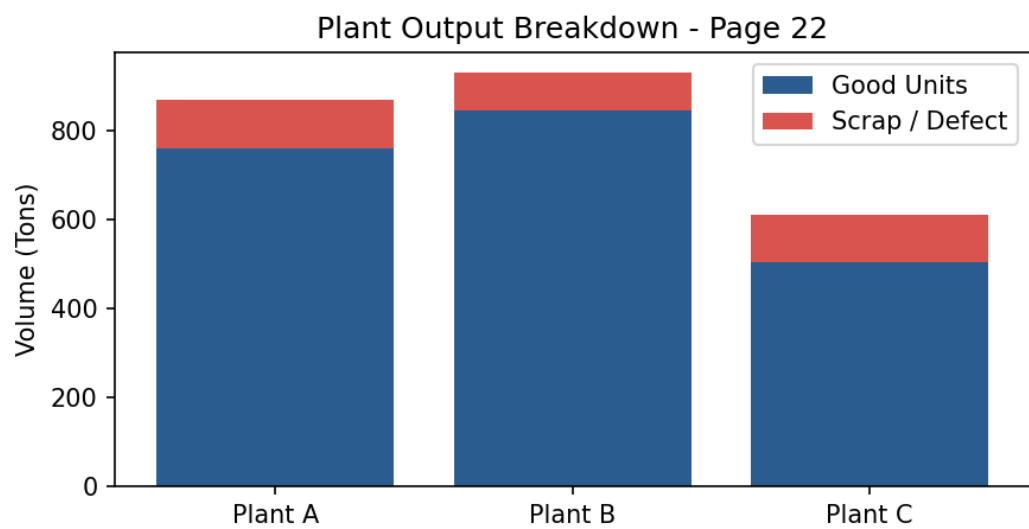
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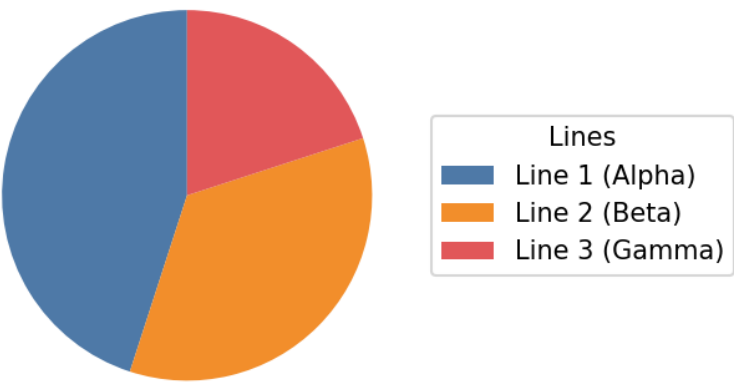
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C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

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ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS